AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-AOH-I15*

Team Name: *WATCH DOGS*

Domain: *Energy,Mobility and Smart Infrastructure(Domain-2)*

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Presentation Outline

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Problem Statement

Campus electricity demand, solar power generation, battery availability, and EV charging requirements fluctuate throughout the day. Without effective coordination, this can lead to peak electricity demand, higher energy costs, underutilization of renewable energy, and charging delays. Our project aims to develop a predictive smart energy management system that forecasts campus electricity demand and optimizes energy allocation and EV charging schedules to improve efficiency, reduce peak loads, and promote sustainable energy use.

Problem Understanding

- [Who experiences this problem?]
  Campus energy managers
  EV and electric bus operators
  Campus administration
  Students and staff
  Electricity grid operators

- [What causes the problem?]
  - Fluctuating campus electricity demand
  - Uncoordinated EV and bus charging
  - Variable solar power generation
  - Limited battery storage capacity
  - Lack of demand forecasting and smart energy scheduling

- [How does it affect users or stakeholders?]
  Stakeholders may face higher electricity costs, peak-load stress, inefficient solar energy use, and delayed EV charging. This can reduce campus energy efficiency and disrupt daily operations.

Existing Gaps & Challenges

- Existing gap or limitation 1:
  Gap 1: Energy demand and renewable power are often managed separately.
- Existing gap or limitation 2:
  Gap 2: EV charging schedules may not account for changing demand, tariffs, and departure times
- Main challenge that remains unsolved:
  Coordinating campus electricity demand, solar generation, battery storage, and EV charging efficiently. Balancing energy costs and peak loads while meeting vehicle charging needs. Developing a predictive system that optimizes energy distribution in real time.

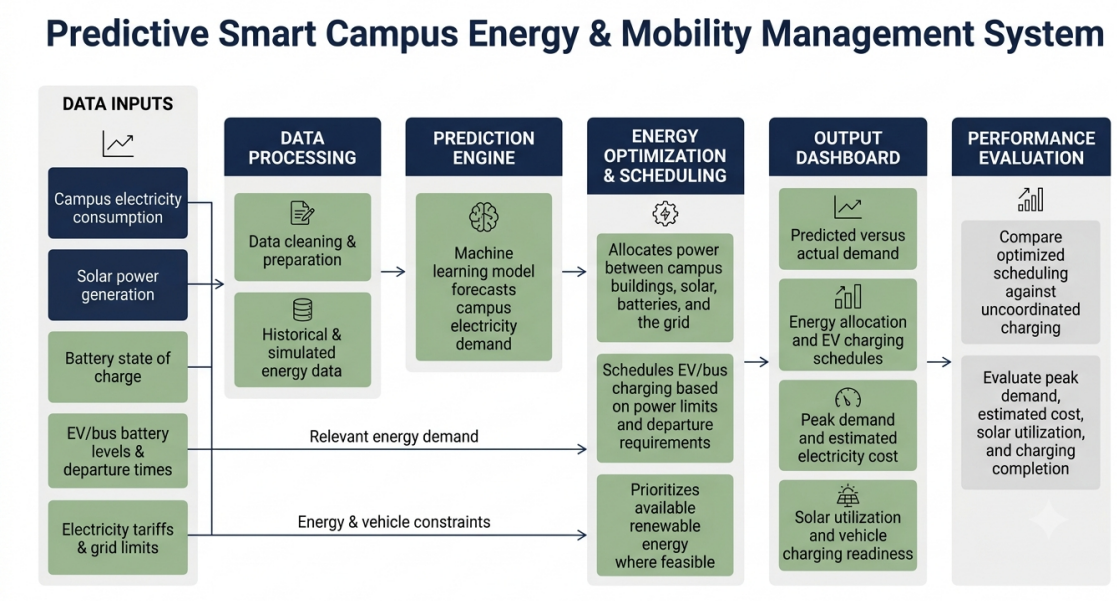
Our Proposed Solution

- Core idea or product:
  An AI-powered smart campus energy management system that predicts electricity demand and optimizes solar power, battery storage, grid usage, and EV charging schedules to reduce peak loads and improve energy efficiency.

- How the solution works at a high level:
  Uses AI to predict campus energy demand and intelligently schedule solar power, battery storage, grid electricity, and EV charging for efficient energy distribution.

- Main value delivered to users:
  Uses AI to predict campus energy demand and intelligently schedule solar power, battery storage, grid electricity, and EV charging for efficient energy distribution

Technical Approach



Tech Stack

Frontend: Streamlit

Backend: Python

Database: CSV files using Pandas for data storage and processing

AI/ML, APIs, or Hardware: Scikit-learn for energy demand prediction; optimization algorithms for energy allocation and EV charging schedules

Innovation & Uniqueness

- What is new or different?
  Combines AI prediction + smart energy scheduling in one system.
  Coordinates solar, batteries, grid power, and EV charging.
  Optimizes energy use based on demand, cost, and departure times.

- Why is this approach suited to the problem?
  Predicts energy demand to anticipate peak loads.
  Optimizes solar, battery, grid power, and EV charging together.
  Enables cost-effective, efficient energy use through simulation and smart scheduling.

- What is the key differentiator?
  Integrates AI-based energy demand forecasting with coordinated EV charging and energy optimization in a single campus-focused system.

Feasibility & Scalability

- **Feasibility:** Feasible using accessible tools like Python, Streamlit, and Scikit-learn with simulated campus energy data. Can be developed within the hackathon timeline without requiring physical hardware.

- **Scalability:** The system can be scaled to larger campuses by adding more buildings, EVs, solar panels, and battery systems. Its modular software design can also be adapted to universities, offices, and smart cities.

- **Risks:**
  Limited real-world data: Use simulated or publicly available datasets.
  Prediction inaccuracies: Train and test models using historical data.
  Scheduling complexity: Start with rule-based optimization and add constraints gradually.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Campus Energy Manager:
  Forecasts campus electricity demand and coordinates solar power, battery storage, and grid electricity to manage peak loads efficiently.

- Use case 2: EV/Bus Operator:
  Plans vehicle charging based on battery levels, available power, and departure times to ensure vehicles are ready when needed.

- Use case 3: Campus Administration:
  Uses the dashboard to monitor energy consumption, estimated costs, peak demand, and solar utilization to support better energy management decisions.

Impacts and Benefits

- Benefit to the target users:
  Lower energy costs and more efficient power management.
  Reliable EV charging and better use of solar energy.

- Social, economic, or environmental impact:
  Reduces energy waste and carbon emissions.
  Supports sustainable, cost-effective campus operations.

- Expected measurable outcome:
  Reduced peak electricity demand and estimated costs.
  Increased solar energy utilization and on-time EV charging.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 7	Target 7.3	Affordable and Clean Energy	Improves energy efficiency and solar power utilization.
SDG 11	Target 11.6	Sustainable Cities and Communities	Promotes efficient campus energy management and reduced environmental impact.

Individual Member Contribution

Member	Contribution or responsibility
Ananya A	AI/ML model for campus energy demand prediction
A Gokulakrishnan	Research, problem statement, stakeholder analysis, and SDG alignment
Thushhar Vinoth	Backend development and energy allocation/EV charging optimization
Sarweshwaran G P	Streamlit dashboard, visualization, testing, and performance evaluation

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